

Nanobody-based CAR NK cells for possible immunotherapy of MICA⁺ tumors

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Abstract

The glycoproteins MICA and MICB are upregulated on the surface of cells undergoing stress, for instance due to (viral) infection or malignant transformation. MICA/B are the ligands for the activating receptor NKG2D, found on cytotoxic immune cells like NK cells, CD8⁺ T cells, and $\gamma\delta$ T cells. Upon engagement of NKG2D, these cells are activated to eradicate the MICA/B-positive targets, assisted by the secretion of cytokines. Nanobodies, or VHHs, are derived from the variable regions of camelid heavy-chain only immunoglobulins. Nanobodies are characterized by their small size, ease of production, stability, and specificity of recognition. We generated nanobodies that recognize membrane-bound MICA with high affinity. Here, we use these nanobodies as building blocks for a chimeric antigen receptor (CAR) to establish VHH-based CAR NK cells. These anti-MICA nanobody-based CAR NK cells recognize and selectively kill MICA-positive tumor cells in vitro and in vivo. We track localization of the VHH-based CAR NK cells to MICA-positive lung metastases by immuno-positron emission tomography imaging.

Keywords: MICA, nanobodies, immunotherapy, CAR NK cells, PET imaging

Significance Statement

MICA is a Class I MHC-related surface glycoprotein, upregulated by virus-infected or malignantly transformed cells. MICA is overexpressed on cancers of hematopoietic and epithelial origin but is absent from healthy cells. We generated nanobodies, the recombinantly expressed variable regions of camelid heavy-chain only immunoglobulins, that recognize MICA with high affinity. We use the nanobodies as building blocks for chimeric antigen receptors (CAR) on NK-92 cells, which recognize and selectively kill MICA⁺ tumor cells in vitro and in vivo. We track the localization of the nanobody-based CAR NK cells to lung metastases of mice by immuno-positron emission tomography imaging. The presence of MICA on many tumor types, and absence from healthy tissue, makes it a promising target for cancer immunotherapy.

Introduction

The MHC class I chain-related proteins A and B (MICA and MICB) are surface glycoproteins that are absent from healthy cells but upregulated on malignantly transformed or otherwise stressed human cells (1). High levels of MICA/B have been reported in cancers of both hematopoietic and epithelial origin (1–6). MICA/B are ligands for the activating receptor NKG2D, found on NK cells, CD8⁺ T cells, and $\gamma\delta$ T cells (7). Upon engagement of MICA/B with NKG2D, these cytotoxic immune cells are activated to eradicate MICA-positive targets (4, 8, 9).

The tumor microenvironment (TME) affects surface expression of the ligands of NKG2D transcriptionally and post-translationally. Surface expression of MICA and MICB on tumor cells can be downregulated through shedding, mediated by proteolytic cleavage involving ADAM-type metalloproteases at the MICA $\alpha 3$ domain (10). Loss of surface-bound MICA renders tumor cells less sensitive to NKG2D-positive NK cells (11, 12).

Nanobodies, also known as VHHs, are the variable heavy-chain fragments of camelid-derived heavy-chain only immunoglobulins (13). Nanobodies are characterized by their small size compared to conventional immunoglobulins (15 kD vs. 150 kD), which allows for excellent tissue penetration, stability, solubility, and ease of production (14–16). Nanobodies are poorly immunogenic and have a short circulatory half-life, making them valuable tools for the construction of positron emission tomography (PET) imaging agents (17–23), nanobody-drug conjugates (24–26), and chimeric antigen receptors in cell-based therapies (27–37).

The latter is based on a cornerstone of immunotherapy known as adoptive cell transfer (ACT), in which immune cells (often the patient's own lymphocytes) are given to a patient as cancer therapy. When using tumor infiltrating lymphocytes expanded ex vivo, this form of treatment is referred to as TIL therapy. In another form of ACT, the patient's T or NK cells are engineered to express a chimeric antigen receptor (CAR) that targets the tumor

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and exerts cytotoxic activity upon binding to the target recognized by the CAR. In such cells, the CAR dictates the antigen specificity of the cell towards a target of choice. The CAR also contains intracellular signaling domains derived from several proteins such as 4-1BB, CD28, and CD3 ζ . These signaling domains activate the CAR cell in response to antigen recognition (38, 39) and trigger cytotoxic activity as well as cytokine release. Typically, the antigen recognition domain of the CAR is based on a single-chain variable fragment (scFv), derived from a full-sized immunoglobulin by connecting the variable regions of the immunoglobulin heavy and light chains by means of a short linker into a single construct. The affinity and specificity of scFvs must be carefully compared to that of the source immunoglobulin to maintain its functional properties. When expressed in mammalian cells, domain swaps can lead to self-aggregation of scFv-based CARs (40–43). The possible immunogenicity of the scFv is a factor to be considered as well. Nanobodies are poorly immunogenic in humans, presumably because of the pronounced homology between camelid and human variable heavy (V_H) chain sequences (44, 45).

The FDA has approved several CAR T-cell therapies for hematopoietic cancers, such as relapsed or refractory B-cell lymphoma or acute lymphatic leukemia, based on CD19 targeting with an scFv, and relapsed or refractory multiple myeloma, based on CARs that target B-cell maturation antigen via either an scFv or a VHH (46–48), the latter with remarkable clinical efficacy. A major reported side effect of CAR T therapy is cytokine release syndrome (CRS), which is systemic inflammation caused by excessive cytokine secretion by the CAR T cells. The cytokines released by NK cells do not induce such inflammation and thus do not cause CRS (49). For these reasons, CAR NK cell therapy is potentially a safer alternative to CAR T-cell therapy.

CAR NK cells can be produced from a variety of sources: from the patient's or a donor's peripheral blood, from a placenta or from umbilical cord blood, existing immortalized NK cell lines (NK-92) or manufactured from induced pluripotent stem cells (iPSC) (31, 50–54). Unlike T cells, NK cells do not pose the risk of GVHD in an allogeneic setting. In fact, NK cells are believed to protect against GVHD in other T-cell-based cancer treatments (55–59). Furthermore, NK cells also allow the inclusion of a wider range of costimulatory domains, using not only traditional intracellular domains derived from CAR T therapies based on CD28, 4-1BB, and CD3 ζ but also NK-specific domains such as CD244, CD137, and NK-Ars (49, 54, 60). If a tumor was to downregulate the CAR's target in an attempt at immune escape, the NK cells might still be effective against the tumor cells because of their intrinsic cytotoxic activity. NK cell-based therapies have entered clinical trials for targeting NY-ESO-1 in synovial carcinoma, myxoid liposarcoma, multiple myeloma, or certain solid tumors. These NK cells are harvested from cord blood and modified with an NY-ESO-1 TCR and IL-15 receptors. Several other preclinical studies with CAR NK cells include CARs that target CD19 and CD20 in B-cell lymphoma and leukemia (31, 61–67), GD2 in neuroblastoma and breast cancer (68–70), and HER2 in breast cancer and other epithelial cancers (71, 72).

Ideally, the target of CAR immune cells is present only on tumor cells and absent from normal tissue, to reduce unwanted off-target effects. Because MICA is expressed primarily on stressed and cancerous cells, MICA is an appealing target for ACT. We have described the production of MICA-targeting nanobodies, VHH-A1 and VHH-H3. These nanobodies recognize the MICA*008 and *009 alleles with nanomolar affinity and recognize surface-disposed MICA on cancer cells (73). Here, we use these anti-MICA nanobodies to establish VHH-based CAR NK cells. We

show that these CAR NK cells recognize and selectively kill MICA-positive tumor cells in vitro and in vivo.

Materials and methods

Cell culture and flow cytometry

The NK-92 cells were obtained from S.K. Dougan (Dana Farber Cancer Institute). NK-92 cells were cultured in complete α MEM (α MEM; no nucleosides, supplemented with 12.5% horse serum, 12.5% fetal bovine serum [FBS], 2 mM L-glutamine, 0.1 M 2-mercaptoethanol, 0.02 mM folic acid, 100 U/mL recombinant IL-2, and 100 U/mL penicillin/streptomycin). B16F10 murine melanoma cells and EL-4 lymphoma cells, and their MICA⁺ transfectants, were obtained from K. Wucherpfennig (Dana Farber Cancer Institute). HEK293T and B16F10 cells were cultured in complete DMEM (DMEM with 4.5 g/L glucose supplemented with 10% FBS and 100 U/mL penicillin/streptomycin). To avoid proteolytic cleavage of membrane-bound MICA, we dissociated adherently grown B16F10 cells from the plate using a 0.5 mM EDTA solution (Gibco). EL-4 cells were cultured in complete RPMI 1640 (RPMI 1640 supplemented with 10% FBS and 100 U/mL penicillin/streptomycin). All cells were cultured to maintain optimal densities and kept in a humidified 5% CO₂ incubator at 37°C. Recognition of surface-disposed MICA by the nanobodies was verified by flow cytometry (73).

Expression of the CAR construct was verified by flow cytometry. Briefly, cells were stained with Cy5-conjugated recombinant extracellular MICA*009 (1 μ g/mL) for 30 min on ice. Cells were washed and viability was determined with LIVE/DEAD™ Fixable Violet Dead Cell Stain Kit (Invitrogen) according to manufacturer's directions. Cells were analyzed on an LSR2 Flow Cytometer (BD Biosciences). Recombinant MICA*009 was produced in-house by transfection of EXPI-293 cells (73).

Mice

C57BL/6J mice and Rag1-deficient mice were purchased from the Jackson Laboratory or bred in-house. Mice were used at 7–12 weeks of age. Experiments were performed in accordance with the guidelines of the Institutional Animal Care and Use Committee (IACUC) of Boston Children's Hospital. Mice were housed under specific pathogen-free conditions in a controlled environment with a 12-light-dark cycle and ad libitum access to standard laboratory chow and water. Health status and welfare of the mice were monitored regularly throughout the study.

Design of the VHH CAR construct and virus production

The gene fragments for the CAR were inserted by “sticky-end” cloning using the BamHI and ClaI restriction enzymes (both from New England Biolabs) into a lentiviral backbone with a mammalian EF-1a promoter lenti-EF1a-IRES-GFP, a gift from David Williams (Boston Children's Hospital) (Fig. S1). For lentiviral production, we transfected HEK-293T cells with 3 μ g of CAR plasmid, 2 μ g of psPAX2 packaging vector, a gift from Didier Trono (Addgene plasmid #12260; <http://n2t.net/addgene:12260>; RRID:Addgene_12260) and 1 μ g of pMD2.G envelope vector, a gift from Didier Trono (Addgene plasmid #12259; <http://n2t.net/addgene:12259>; RRID:Addgene_12259) in 500 μ L of serum-free DMEM (Gibco). This DNA mixture was added to 100 μ L of serum-free DMEM at a 6:1 ratio of FuGENE6 transfection reagent (Promega) and incubated for 15–30 min at room temperature. The mixture was then added to ~70% confluent HEK293T cells

grown in 10 mL of complete DMEM. The medium was replaced ~16 h after transfection. Lentivirus was harvested 24 and 48 h after the medium change, the media combined, and concentrated by centrifugation at 45,000 $\times$ g for 2 h at 4°C.

Lentiviral transduction and selection of transduced NK cells

NK cells were transduced by centrifugal inoculation. Briefly, 1×10^5 NK cells were added to a well of a 6-well plate, at a 1:3 ratio of concentrated viral supernatant and complete α MEM. Polybrene infection agent (Sigma-Aldrich) (8 μ g/mL) was added to improve transduction. BX795 (1.5 μ M), IL-2 (500 IU/mL), and IL-12 (20 ng/mL, all from PeproTech) were added for optimal cell viability. Cells were centrifuged at 2000 $\times$ g at 30°C, for 90 min, re-suspended, and incubated in the viral culture medium at 37°C in a humidified 5% CO₂ atmosphere for 4 h. Centrifugation was repeated at 2000 $\times$ g at 30°C, for 60 min. The medium was then replaced with complete α MEM.

Activation of CAR NK cells by coculture with MICA-expressing B16F10 cells

WT or MICA-expressing B16F10 or EL-4 cells were incubated at 25,000 cells per well of a 96-well plate together with CAR NK cells at various effector to target ratios in a total volume of 100 μ L (1:1 ratio of complete DMEM and complete α MEM) at 37°C in a humidified 5% CO₂ atmosphere. After 24 h, the IFN γ concentration in the medium was determined by ELISA, using the human IFN γ matched antibody pair (ThermoFisher scientific) according to the manufacturer's instructions. Cell death was determined with a lactate dehydrogenase (LDH) cytotoxicity assay (Abcam, Ab65393) performed according to the manufacturer's instructions.

CAR NK treatment in vivo

Rag1^{-/-} mice were subcutaneously injected in the right flank with 4×10^5 B16F10 MICA⁺ cells in PBS. Retro-orbital injections of CAR NK were started on day 3 and treatment injections were given twice weekly at $5\text{--}10 \times 10^6$ cells per injection. Tumor size was measured by calipers and tumor volume was calculated using the formula ($V = 0.5 \times L \times W^2$). Mice were sacrificed once the tumor volume reached 2000 mm³ or when ulcerations were observed.

PET-CT imaging

To create the imaging agents for PET-CT imaging, we ligated GGG-DFO-Azide to VHH₁₈₈ by sortase-mediated transpeptidation. For sortase reactions, the nanobody was incubated with a 10-fold molar excess of GGG-DFO-Azide and 25 μ M Sortase 7 M (74) overnight at 4°C. Reaction mixtures were depleted of unreacted VHH and Sortase, both containing a C-terminal (His)₆-tag on a NiNTA matrix and elimination of free nucleophile by desalting on a PD-10 column (Cytiva), eluting in fractions of 500 μ L PBS. We selected and combined fractions 6, 7, and 8 (Fig. S2A). To extend the half-life of the nanobody in vivo, the nanobodies were PEGylated by incubation with a 10-fold molar excess of DBCO-PEG20 overnight at 4°C. The reaction was cleaned with a PD-10 column (Cytiva), eluting in fractions of 500 μ L PBS. We selected and combined fractions 6, 7, and 8 (Fig. S2B). ⁸⁹Zr was ordered from the UW-Madison Cyclotron Lab (University of Wisconsin-Madison, USA) and neutralized to a pH of 7.4 by addition of 2 M Na₂CO₃ and 1 M HEPES. Nanobodies were labeled with ⁸⁹Zr by DFO-mediated chelation in chelexed PBS and excess, unbound ⁸⁹Zr was removed by desalting on a PD-10 column and eluted in fractions of 600 μ L chelexed PBS (Fig. S3). Radioactivity

of the individual fractions was determined and ~60 μ Ci VHH₁₈₈-PEG20-⁸⁹Zr was injected per mouse via retro-orbital injection. Mice were anesthetized using 2% isoflurane in O₂ at a flow rate of 1 liter per minute. PET/CT scans were obtained 1-, 24-, 48-, and 72-h postinjections on a G8 PET-CT small-animal scanner (PerkinElmer). Each scan had a PET acquisition time of 10 min, followed by a CT-scan for 1.5 min. PET images were processed and analyzed using VivoQuant software.

Statistical analysis

All statistical analysis was performed with GraphPad Prism 8. Flow cytometry data were analyzed with FlowJo (v10.8.1 and v10.9.0).

Results

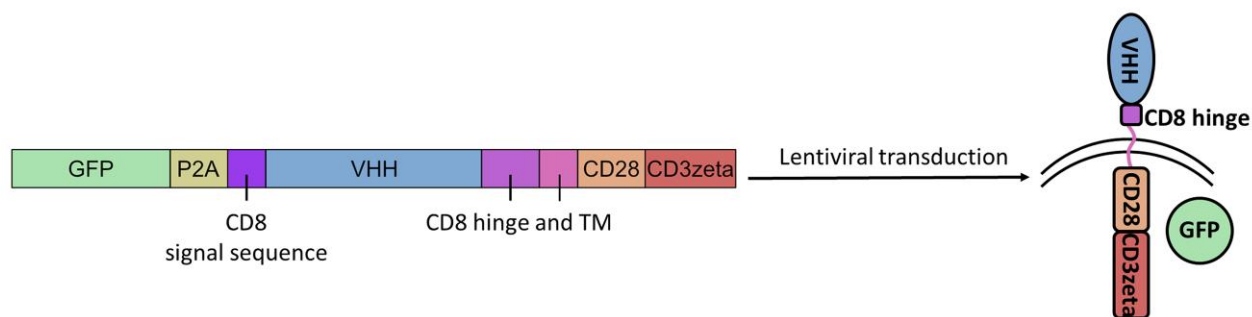
Transduction with lentiviral anti-MICA VHH-based CAR constructs yielded stable CAR NK cells

The design of CAR NK cells was based on previously described VHH-based CAR T cells (29) (Fig. 1A). We designed a GBlockTM Gene fragment that encodes GFP, followed by a P2A proteolytic processing site to separate the GFP portion from the CAR itself. The CAR antigen recognition domain is encoded by the MICA-specific VHHs A1 or H3, followed by the CD8 hinge and CD8 transmembrane segment and the cytoplasmic signaling and costimulatory domains of CD28 and CD3 ζ . The characterization of the anti-MICA VHHs has been described (73). Cells bearing these VHH-based CARs will be referred to as A1 and H3 CAR NK cells. As a control, we transduced NK cells with a lentiviral vector containing only a GFP cassette, referred to as empty vector ("EV"). For the NK cells, we observed a low transduction efficiency between 0.3 and 5% (Fig. 1B). We therefore sorted the GFP⁺ cells to establish a stably transduced cell line. We included a PE channel to eliminate dead auto-fluorescent cells that show up in the PE channel. Flow cytometry performed on the H3 CAR NK cells shows the presence of two distinct populations of GFP⁺ cells, attributed to the combination of two cell lines transduced on separate days, done to obtain adequate numbers of cells (Fig. 1C). To verify CAR surface expression, we stained the CAR NK cells with Cy5-conjugated MICA protein, which binds to the extracellularly exposed nanobodies that are part of the CAR. Flow cytometry produced a clear signal in the Cy5 channel for the A1 CAR NK cells, showing surface expression of the VHH-A1-based CAR. We saw a weak signal in the Cy5 channel for the H3 CAR NK cells, indicating weaker expression of the VHH-H3-based CAR (Fig. 1D). For the H3 CAR NK cells, both the GFP^{hi} and the GFP^{lo} population show the same signal in the Cy5 channel (Fig. S4). Immunoblots prepared with anti-CD3 ζ antibodies (signal at 40 kD in A1 and H3 CAR NK cells) showed expression of the CAR. The GFP polypeptide produced by the Lenti-EF1 α A1 and H3 NK cell lysates migrates slightly higher on SDS-PAGE than the GFP produced by the Lenti-EF1 α EV, attributable to the continued presence of the P2A peptide sequence downstream of the GFP in the CAR construct (Fig. 1E). Gating for flow cell sorting and flow cytometry is shown in Figs S5–S7.

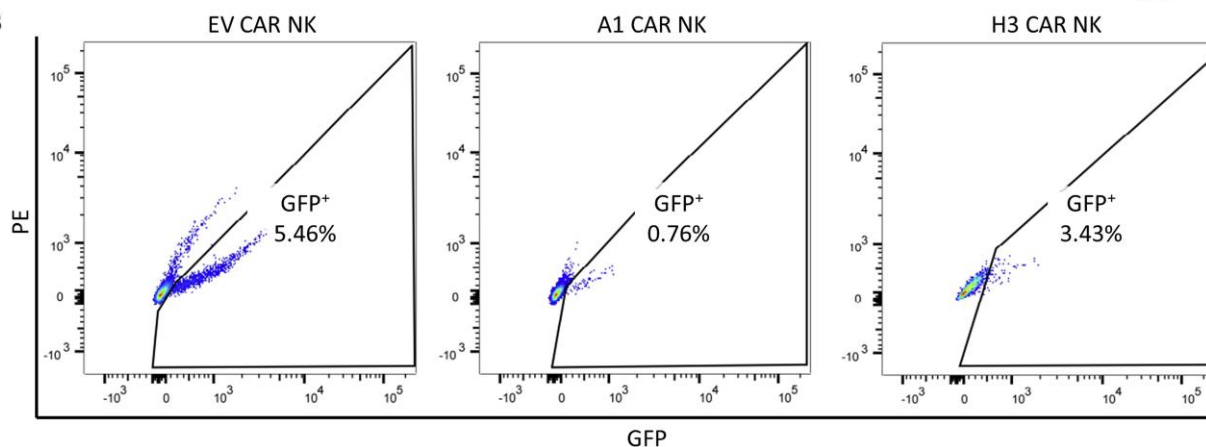
MICA-expressing tumor cells activate A1 and H3 CAR NK cells and elicit cytotoxicity

Because mice lack a protein homologous to human MICA, we used MICA transfectants of the mouse-derived B16F10 melanoma (MICA*009) and EL-4 T-cell lymphoma (MICA*008) lines. We

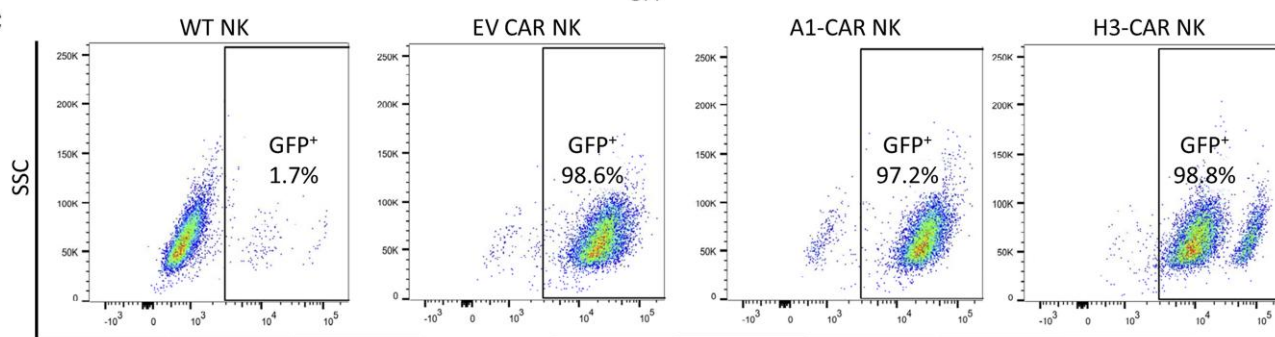
A



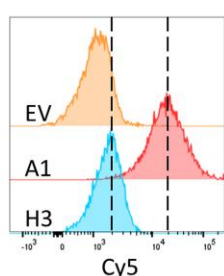
B



C



D



	MFI
EV CAR NK	1268
A1 CAR NK	24873
H3 CAR NK	1877

E

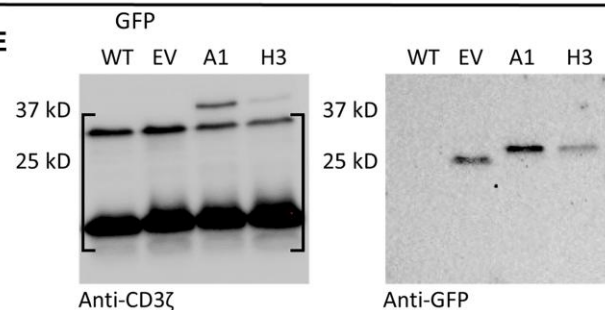


Fig. 1. The production and establishment of stable CAR NK cells. A) Schematic overview of the CAR construct as transduced into NK cells. We used a lentiviral backbone with a mammalian EF1a promoter and incorporated the sequences of either VHH A1 or VHH H3, the costimulatory and activation signals of CD28 and CD3 ζ , and GFP separated from the rest of the construct by a P2A peptide cleavage signal. Construct maps for A1 CAR NK, H3 CAR NK, and EV CAR NK are shown in Fig. S1. B) After transduction, GFP positive cells were deemed to be transduced successfully, and therefore sorted by FACS. We included a PE channel to eliminate dead, auto-fluorescent cells. Gating shown in Fig. S5. C) We continually monitored GFP expression in the CAR NK cells and found a stable expression after at least 4 months in culture, with a GFP positive population of 98.6% for the EV NK cells, 97.2% for the A1 CAR NK cells, and 98.8% for H3 CAR NK cells. Gating shown in Fig. S6. D) To verify CAR surface expression by flow cytometry, we stained the CAR NK cells with Cy5-conjugated recombinant MICA protein. We observed a clear signal in the Cy5 channel for the A1 CAR NK cells, thus establishing surface expression of VHH-A1 as part of the CAR. We saw a weak signal in the Cy5 channel for the H3 CAR NK cells, indicating weaker expression of the VHH-H3-bearing CAR. Gating is shown in Fig. S7. E) Immunoblots prepared with anti-CD3 ζ antibodies showed expression of the actual CAR portion of the viral vector with a signal at 40 kD in A1 and H3 CAR NK cells and an absence of signal in WT and EV CAR NK cells. The GFP polypeptide produced by the Lenti-EF1a A1 and H3 NK cell lysates runs slightly higher than that of GFP produced by the Lenti-EF1a EV cell lysate, which we attribute to the presence of the P2A peptide sequence downstream of the GFP in the CAR construct. The weaker signal observed for the H3 CAR NK cells matches that observed by flow cytometry. The signal in brackets is nonspecific.

incubated A1 and H3 CAR NK cells with WT or MICA⁺ B16F10 cells or with WT or MICA⁺ EL-4 cells. We cocultured effector cells and target cells at different ratios ([1:1], [0.2:1], and [0.1:1]), keeping the number of target cells constant and varying the number of effector cells. For the A1 and H3 CAR NK cells, at all [E:T] ratios, we observed a significant increase in cell death of MICA⁺ cells as measured by LDH release. Coculture of MICA⁺ cells with EV CAR NK cells showed no such increase in cytotoxicity. No cytotoxicity was observed when coculturing WT B16F10 cells with the A1 or H3 CAR NK cells. We observed a significant increase in cytotoxicity of EL-4 MICA⁺ cells cocultured with A1 CAR NK cells or H3 CAR NK cells when compared to cytotoxicity exerted by EV CAR NK cells. No significant increase in cytotoxicity was observed in WT EL-4 cells cocultured with EV, A1, or H3 CAR NK cells (Fig. 2A). To relate this cytotoxicity to activation of CAR NK cells, we measured IFN γ secretion by ELISA. Upon coculture with B16F10 MICA⁺ we observed an increase in IFN γ secretion for the A1 and H3 CAR NK cells, but not for the EV CAR NK cells. CAR NK cells cocultured with B16F10 WT cells showed no such increase. We observed an increase in IFN γ expression when A1 CAR NK cells were cocultured with EL-4 MICA⁺ cells at [1:1], but not for any of the other conditions, despite the observed significant difference in cytotoxicity (Fig. 2B). We attribute this to the fact that we observed lower MICA expression levels on the surface of the EL-4 MICA⁺ cells compared to the B16F10 MICA⁺ cells (73). Furthermore, EL-4 cells are suspension cells, while B16F10 cells are adherent, potentially facilitating an interaction with NK cells. These results indicate the possibility of CAR NK cells to treat MICA positive tumors in vivo. We estimated IFN γ production in pg/mL by extrapolation from the standard curve (Fig. 2C).

MICA⁺ tumor-bearing mice treated with A1 CAR NK cells show reduced tumor growth and increased survival probability

We inoculated RAG1^{-/-} mice with 4×10^5 B16F10 MICA⁺ cells via subcutaneous injection. Twice weekly, we treated the mice with EV CAR NK cells ($n = 3$) or A1 CAR NK cells ($n = 7$) by retro-orbital injection for a total of five injections and followed tumor growth by caliper measurements (Fig. 3A). Mice treated with A1 CAR NK cells show a significant delay in tumor growth compared to mice treated with EV CAR NK cells ($P = 0.0075$) (Fig. 3B). Furthermore, mice treated with A1 CAR NK cells showed a significant increase in survival probability ($P = 0.0011$) (Fig. 3C). Of the mice treated with EV CAR NK cells, one mouse was euthanized on day 17 because of severe ulcerations. Two mice were euthanized on days 17 and 18 when the tumor volume exceeded 2,000 mm³. Of the mice treated with A1 CAR NK cells, one mouse was euthanized on day 20 and one on day 21, both because of severe ulcerations. All other mice were euthanized when the tumor volume exceeded 2,000 mm³. These results support the possibility of nanobody-based CAR NK cells to treat MICA⁺ tumors.

Immuno-PET traces CAR NK cells to MICA⁺ tumors

To track the localization of CAR NK cells to MICA⁺ tumors, we inoculated C57/B6 mice with B16F10 MICA⁺ tumors by tail vein injection and allowed lung metastases to form for 15 days. We used PEGylated ⁸⁹Zr-labeled VHH₁₈₈ (Fig. S2), a nanobody that targets the human transferrin receptor on the NK cells, which are of human origin (Fig. S8) for immuno-PET imaging. VHH₁₈₈ does not recognize the mouse transferrin receptor (75). On day 1 of imaging, mice were injected with either EV CAR NK cells or A1 CAR NK cells (both 5×10^6 cells/mouse) into one retro-orbital

cavity, and ⁸⁹Zr-labeled PEGylated VHH₁₈₈ (60 μ Ci/mouse) into the other retro-orbital cavity. We were able to localize the CAR NK cells to the lungs of the mice that received B16F10 MICA⁺ tumors up to 72 h after NK cell injection (Fig. 4A). We observed slightly more positive nodules in the lungs of the mice that received A1 CAR NK cells compared to EV CAR NK cells at 72 h post-injection, although it is difficult to draw a meaningful conclusion based on these small differences, seen only at the 72-h timepoint. Possibly the EV CAR NK are cleared from the circulation more rapidly, due to their inability to bind to a MICA target. Upon dissection of the lungs, we saw clear metastatic lesions on the surface of the lungs (Fig. 4B). The resolution of the PET images used is ~ 1.4 mm, which makes it difficult to visualize positive signal in greater detail.

Discussion

MICA and MICB are Class I MHC-related proteins expressed on stressed and cancerous cells. Their presence can serve not only as a diagnostic marker of malignancies but also as a possible target for therapy. While typical immunoglobulins exert their functional properties through Fc effector functions, their size compromises efficient tissue penetration. Nanobodies offer an appealing alternative to immunoglobulins for the purpose of launching an immune attack on MICA-positive tumors. We previously produced high affinity nanobodies, A1 and H3, that recognize the extracellular portion of MICA, alleles *008 and *009, on the surface of MICA⁺ B16F10 melanoma cells and MICA⁺ EL-4-T-cell lymphoma cells.

ACT is widely explored as a possible cancer therapy. The success of VHH-based CAR T cells in tumor treatment has been recorded (29), with the first VHH-based CAR T-cell therapy (Carvykti) approved for treatment of relapsed or refractory multiple myeloma (76). Despite the success of CAR T cells, significant drawbacks and side-effects deserve consideration. T cells are often sourced from the patient's own peripheral blood and require expansion ex vivo after modification. CAR NK cells can be obtained from a wider range of sources, such as the patient's or a donor's peripheral blood, from placenta or umbilical cord blood, existing immortalized NK cell lines (NK-92) or manufactured from induced pluripotent stem cells (iPSC) (31, 50–54).

A major reported side effect of CAR T therapy is CRS, a systemic inflammation caused by excessive secretion of cytokines such as IL-2 and IL-6 released by the CAR T cells. Other immune cells may respond to cytokines produced by the CAR T cells and also contribute to pathology. Most of the cytokines released by NK cells (IL-3 and TNF- α) do not induce such inflammation, and thus are less likely to cause CRS (49), although CRS due to overexpression of IFN γ has been reported in a patient receiving CAR NK cells, thus careful monitoring is still required (77). For these reasons, CAR NK cell therapy is potentially a more effective and safer alternative to CAR T-cell therapy.

Here, we developed VHH_{MICA}-based CAR NK cells that target and selectively kill MICA⁺ B16F10 and MICA⁺ EL-4 cells in vitro. Immuno-PET shows that the A1 CAR NK cells localize to the lungs of mice bearing MICA⁺ B16F10 lung metastases. We see such localization until 72 h postinjection. The CAR NK cells are also cytotoxic towards MICA⁺ B16F10 cells in vivo. MICA⁺ B16F10 tumor-bearing mice treated with A1 CAR NK cells show a significant reduction in the rate of tumor growth and increase in overall survival compared to mice treated with EV CAR NK cells.

We recognize the limitation of using mouse-derived cancer cells that have been rendered MICA-positive by transfection. A major constraint is the availability of patient-derived cancer cell

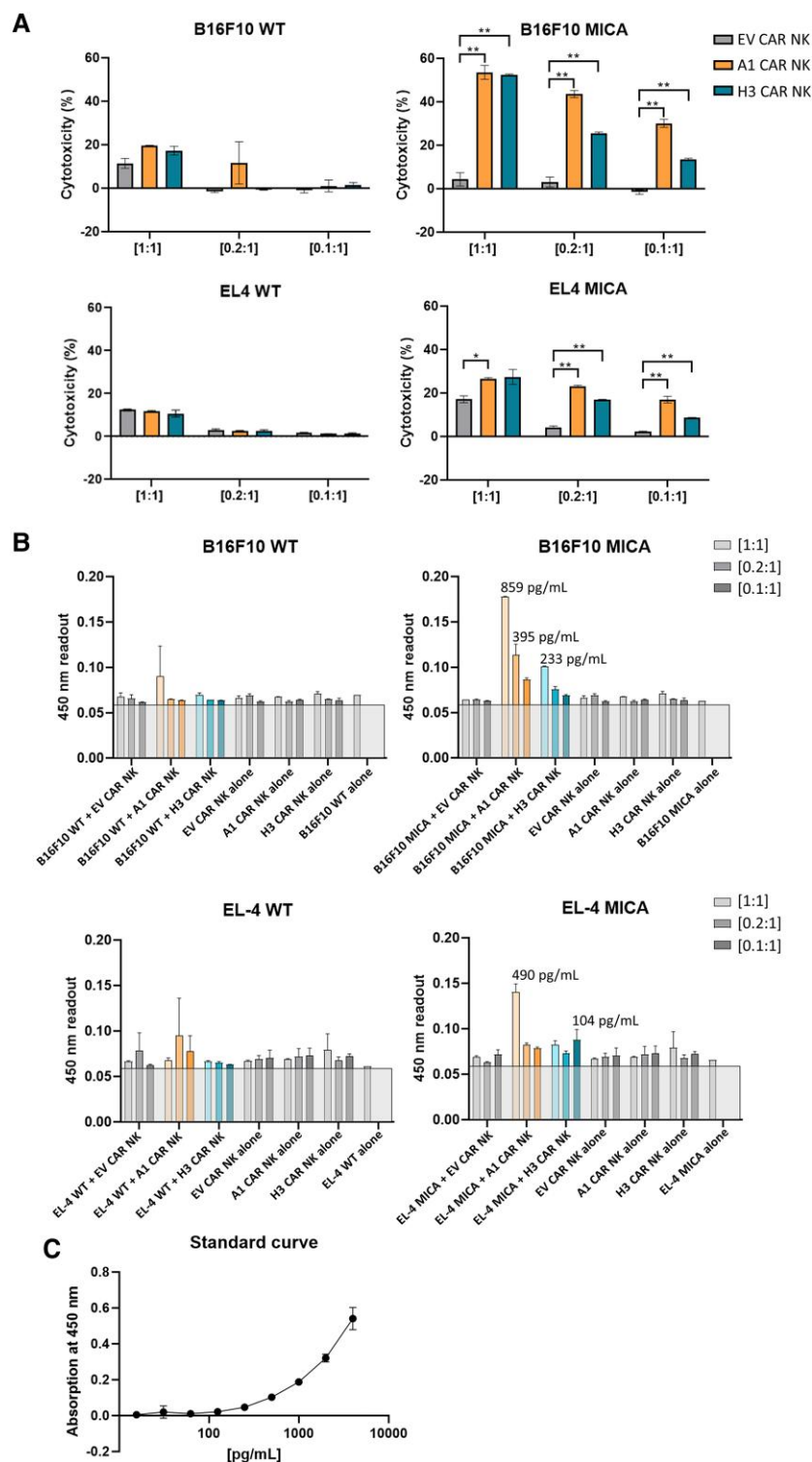


Fig. 2. In vitro cytotoxicity of CAR NK cells cocultured with MICA⁺ targets. We incubated EV, A1, and H3 CAR NK cells with WT B16F10 or EL-4 cells, or B16F10 or EL-4 cells that stably express MICA. We incubated at effector to target ratios [E:T] of [1:1], [0.2:1], or [0.1:1], keeping the number of target cells consistent and varying the number of effector cells. A) Twenty-four hours after coculture, cytotoxicity was determined by measuring LDH release in the medium. Cytotoxicity percentages were normalized to target cells without CAR NK co-culture as a background control (0% cell death), or lysed target cells as a high control (100% cell death). We observed a significant increase in cytotoxicity of B16F10 MICA⁺ cells cocultured with A1 CAR NK cells and H3 CAR NK cells compared to EV CAR NK cells at all [E:T] ([1:1] $P = 0.004$ for A1; $P = 0.002$ for H3, [0.2:1] $P = 0.0024$ for A1; $P = 0.0055$ for H3 and [0.1:1] $P = 0.0022$ for A1; $P = 0.0027$ for H3). EL-4 MICA⁺ cells showed a lower overall cytotoxicity of 15–30% when cocultured with A1 and H3 CAR NK cells. We observed a significant increase in cytotoxicity of EL-4 MICA⁺ cells cocultured with A1 CAR NK cells or H3 CAR NK cells compared to EV CAR NK cells at most [E:T] ([1:1] $P = 0.015$ for A1; $P = 0.061$ for H3, [0.2:1] $P = 0.0011$ for A1; $P = 0.0016$ for H3, [0.1:1] $P = 0.0055$ for A1; $P = 0.0014$ for H3). No significant increase in cytotoxicity was observed when WT cells were cocultured with EV, A1, or H3 CAR NK cells. B) After 24 h of coculture, the concentration of IFN γ in the medium was determined by sandwich ELISA with a matched human IFN γ antibody pair. We observed a significant increase in IFN γ secretion in the A1 and H3 CAR NK cells, but not in EV CAR NK cells, when cocultured with B16F10 MICA⁺. We show the raw values of the ELISA plate read-out at 450 nm and the estimated IFN γ production in pg/mL by extrapolation from the standard curve (C).

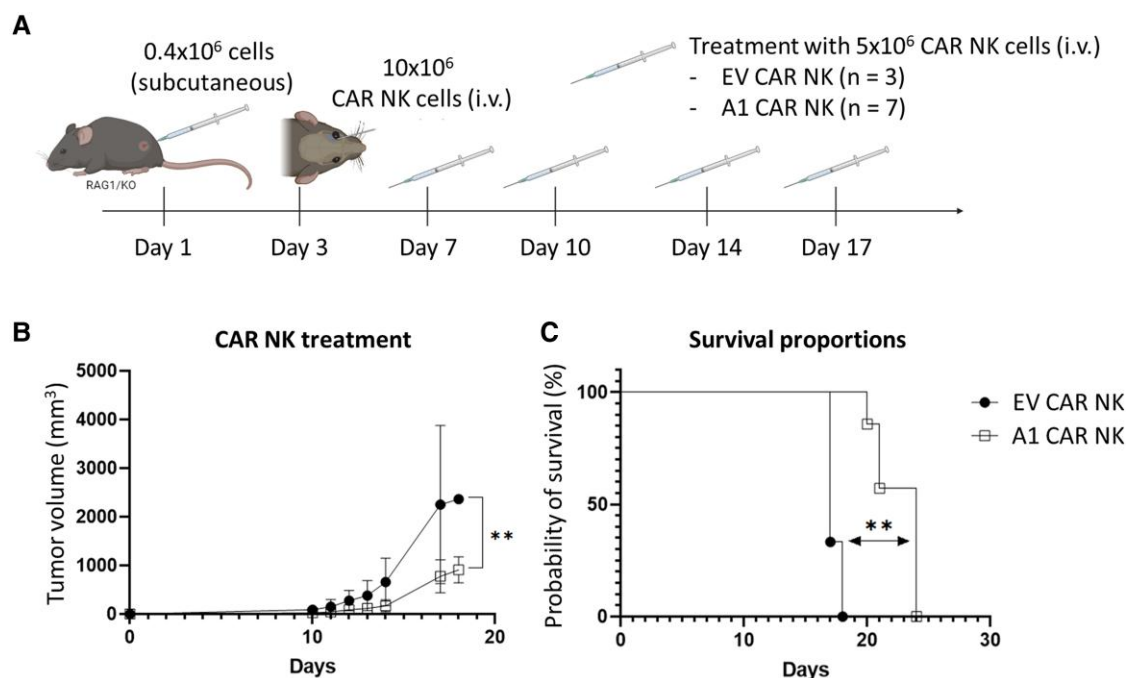


Fig. 3. In vivo cytotoxicity of CAR NK cells in MICA⁺ tumor-bearing mice. A) We challenged RAG1/KO mice with B16F10 MICA cells by subcutaneous injection. On day 3 post tumor graft, we treated the mice with a retro-orbital dose of 10×10^6 CAR NK cells and reduced the dose to 5×10^6 cells per mouse for the treatments twice weekly thereafter. B) Mice treated with A1 CAR NK cells show a significant delay in tumor growth compared to mice treated with EV CAR NK cells ($P = 0.0075$) (calculated with two-way ANOVA). C) Mice treated with A1 CAR NK cells showed a significant increase in survival probability ($P = 0.0011$) (calculated with the Mantel-Cox test).

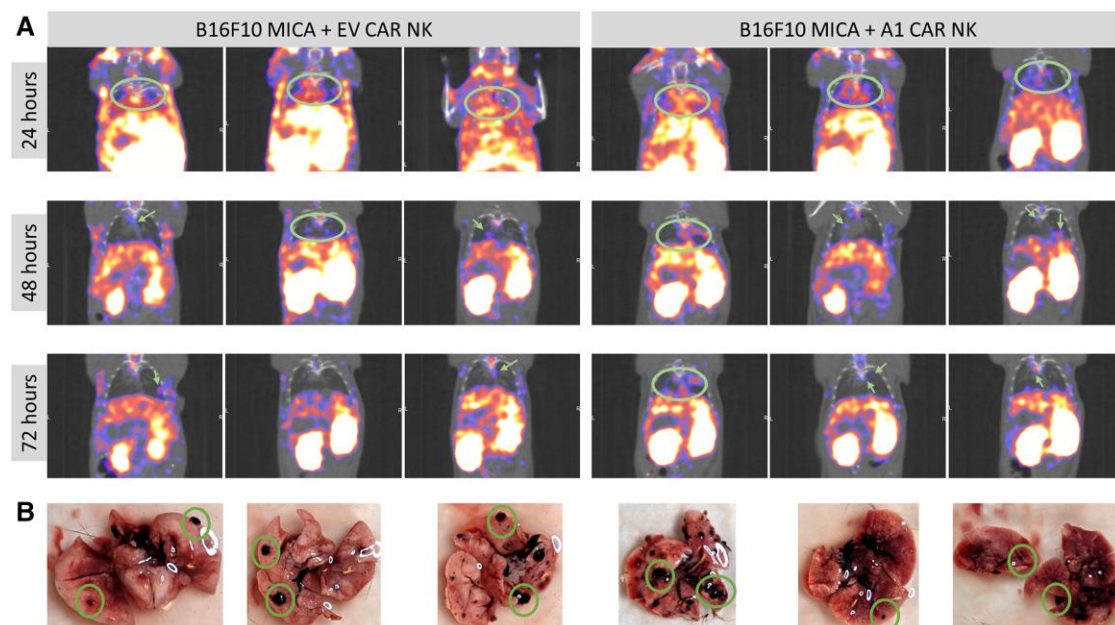


Fig. 4. PET imaging to track localization of CAR NK cells to MICA⁺ lung metastases. A) We injected C57/Bl6 mice with B16F10 MICA⁺ cells by tail vein injection and allowed metastases to form for 15 days. On day 1 of imaging, EV and A1 CAR NK cells and ⁸⁹Zr-VHH₁₈₈ were injected retro-orbitally. We were able to localize the A1 CAR NK cells to the lungs of the mice that received B16F10 MICA⁺ tumors up to 72 h after NK cell injection (circles and arrows). Little if any signal remained at the 72-h timepoint for mice that received EV CAR NK cells. Regardless of the type of CAR NK cells transferred, strong PET signals were seen in the liver. In grayscale: CT density in Hounsfield units (HU), in color: PET intensity in Bq/mL. Coronal sections through the lungs are shown. For full images and intensity scales, see Fig. S9. For 3D videos, see [10.6084/m9.figshare.25471795](https://doi.org/10.6084/m9.figshare.25471795). B) We dissected the lungs of each mouse and observed metastases on the lung surface (several highlighted by circles).

lines that not only express the correct alleles of MICA but that are also suitable for transplantation. Using such lines for engraftment of immunocompetent mice poses a risk of a possible xenogeneic response independent of MICA expression and thus requires the use of immunodeficient recipients.

Although MICA is generally absent from healthy tissue, expression of MICA is seen in gut epithelium, although primarily intracellularly (78, 79). Since gut epithelia are capable of rapid repair, this risk may be manageable, should MICA-specific CAR NK cells indeed attack gut epithelia. Nevertheless, since mice do not

possess a MICA homolog, the use of MICA-transgenic mice (80) might allow an assessment of any “off-tumor, on-target” effects when using MICA-targeting CAR NK cells.

The genetic instability of NK-92 cells requires their irradiation prior to infusion in a patient, which impairs their proliferation and limits their persistence in vivo (81). Patient-specific induced pluripotent stem cell-derived NK cells (iPSC-NKs) may be better for CAR T and CAR NK cell therapy. iPSC-NK cells express the CD16 Fc receptor and are thus capable of antibody-dependent cellular cytotoxicity (52, 82–84).

Despite excellent results in the treatment of certain hematological cancers, the efficiency of CAR treatment in solid tumors remains poor. The CARs used in this study are based on what is referred to as a “second-generation” CAR, which includes a CD3 ζ signaling domain and CD28 costimulatory domain. Possible improvements to this CAR design include the addition of a cytokine autostimulation domain, such as IL-15 (85, 86). Third- and fourth-generation CARs employ additional costimulatory domains such as CD27 or STAT3/5 binding motifs. Other modifications, such as enhanced CD28 signaling domains or the inclusion of ITAMs 2 and 3 in CD3 ζ may increase the stability of CAR cells and thus show better tumor control in vivo (87).

The tumor microenvironment often shows deposition of extracellular matrix (ECM) components and may cause encapsulation of a solid tumor that limits access to the tumor for CAR T or CAR NK cells. Instead of CAR T or CAR NK cells, MICA-specific CAR macrophages might help degrade the ECM by secretion of proteases and improve the outcome of immunotherapy (88–93). Because VHH A1 and VHH H3 recognize different epitopes on MICA (73), we could use H3-based CAR macrophages to help degrade the ECM and attract A1-based CAR NK cells to aid in tumor-specific cytotoxicity, without the different cell types competing for binding to MICA.

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Supplementary Material

Supplementary material is available at PNAS Nexus online.

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Author Contributions

The authors confirm their contribution to the article as follows: E.R.V. and H.L.P. designed the study and supervised data collection. E.R.V. and W.v.K. designed and produced the CAR NK cells. E.R.V. and A.K. performed research with the CAR NK cells. T.B. provided guidance on PET imaging and the work with the anti transferrin receptor nanobody. E.R.V. and H.L.P. wrote the article.

All authors reviewed the results and approved the final version of the manuscript.

Data Availability

All data are included in the manuscript and supporting information. 3D videos of PET/CT imaging available at [10.6084/m9.figshare.25471795](https://doi.org/10.6084/m9.figshare.25471795) Raw data files for in vitro and in vivo cytotoxicity assays can be available at [10.6084/m9.figshare.25733889](https://doi.org/10.6084/m9.figshare.25733889).

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